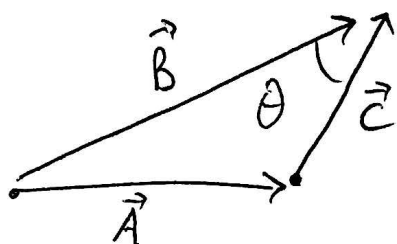


Problem 1



$$\vec{A}^2 = (\vec{B} - \vec{C})^2$$

$$\underbrace{A \cdot A \cdot \cos(0)}_{A^2} = \underbrace{\vec{B}^2}_{B^2} + \underbrace{\vec{C}^2}_{C^2} - 2 \vec{B} \cdot \vec{C}$$

$$A^2 = B^2 + C^2 - 2BC \cos(\theta)$$

$$\Rightarrow A^2 = B^2 + C^2 - 2BC \cdot \cos \theta$$

ANSWER 1 ~~XX~~

Problem 2

$$(\vec{A} \times \vec{B}) \cdot (\vec{A} \times \vec{B}) = (AB)^2 - (\vec{A} \cdot \vec{B})^2$$

Let $\vec{A} \times \vec{B} = \vec{C}$ then ~~just~~

$$\vec{C} \cdot \vec{C} = |\vec{C}| \cdot |\vec{C}| \cdot \cos 0 = C^2$$

$$|\vec{C}| = |\vec{A}| |\vec{B}| \sin \theta = A \cdot B \cdot \sin \theta$$

left side:

$$(A \cdot B \cdot \sin \theta)^2 \stackrel{?}{=} (AB)^2 - (\vec{A} \cdot \vec{B})^2$$

$$\downarrow$$
$$A^2 \cdot B^2 \cdot \sin^2 \theta$$

$$\sin^2 \theta = 1 - \cos^2 \theta$$

$$A^2 \cdot B^2 \cdot (1 - \cos^2 \theta) = A^2 \cdot B^2 - A^2 \cdot B^2 \cdot \cos^2 \theta$$

$$\underbrace{A \cdot B \cdot \cos \theta}_{\vec{A} \cdot \vec{B}} \cdot \underbrace{A \cdot B \cdot \cos \theta}_{\vec{A} \cdot \vec{B}}$$

left side becomes:

$$A^2 B^2 - (\vec{A} \cdot \vec{B})(\vec{A} \cdot \vec{B})$$

$$= A^2 B^2 - (\vec{A} \cdot \vec{B})^2 = A^2 B^2 - (\vec{A} \cdot \vec{B})^2$$

ANSWER 2

Problem 3

$$\text{Let } \vec{D} = \vec{B} \times \vec{C} = \overbrace{(B_y C_z - B_z C_y)}^{D_x}, \overbrace{(B_z C_x - B_x C_z)}^{D_y}, \overbrace{(B_x C_y - B_y C_x)}^{D_z}$$

$$\text{then } (\vec{A} \times \vec{D})_x = A_y D_z - A_z D_y$$

$$= A_y (B_x C_y - B_y C_x) - A_z (B_z C_x - B_x C_z)$$

$$= \underbrace{A_y B_x C_y}_{\text{left side}} - \underbrace{A_y B_y C_x}_{\text{right side}} - \underbrace{A_z B_z C_x}_{\text{right side}} + \underbrace{A_z B_x C_z}_{\text{left side}}$$

$$B_x \cdot (A_y C_y + A_z C_z) - C_x (A_y B_y + A_z B_z) \rightarrow \text{left side}$$

$$\vec{B}_x (A_x C_x + A_y C_y + A_z C_z) - C_x (A_x B_x + A_y B_y + A_z B_z) \rightarrow \text{right side}$$

$$B_x \cancel{A_x C_x} + \underbrace{B_x (A_y C_y + A_z C_z)}_{\text{left side}} - C_x \cancel{A_x B_x} - \underbrace{C_x (A_y B_y + A_z B_z)}_{\text{right side}}$$

left side:

$$B_x (A_y C_y + A_z C_z) - C_x (A_y B_y + A_z B_z) = B_x (A_y C_y + A_z C_z) - C_x (A_y B_y + A_z B_z)$$

ANSWER 3

Problem 4

Multiplying with $\cdot (\vec{B} \times \vec{C})$

$$\vec{D} \cdot (\vec{B} \times \vec{C}) = a \vec{A} \cdot (\vec{B} \times \vec{C}) + \underbrace{b \vec{B} \cdot (\vec{B} \times \vec{C})}_0 + \underbrace{c \vec{C} \cdot (\vec{B} \times \vec{C})}_0$$

$\vec{B} \perp (\vec{B} \times \vec{C}) \quad \vec{C} \perp (\vec{B} \times \vec{C})$

$$\vec{D} \cdot (\vec{B} \times \vec{C}) = a \vec{A} \cdot (\vec{B} \times \vec{C})$$

$a = \frac{\vec{A} \cdot (\vec{B} \times \vec{C})}{\vec{D} \cdot (\vec{B} \times \vec{C})}$	ANSWER 4.1
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for b. multiplying with $\cdot (\vec{A} \times \vec{C})$

$$\vec{D} \cdot (\vec{A} \times \vec{C}) = \underbrace{a \vec{A} \cdot (\vec{A} \times \vec{C})}_0 + b \vec{B} \cdot (\vec{A} \times \vec{C}) + \underbrace{c \vec{C} \cdot (\vec{A} \times \vec{C})}_0$$

$$\vec{D} \cdot (\vec{A} \times \vec{C}) = b \vec{B} \cdot (\vec{A} \times \vec{C})$$

$$-\vec{A} \cdot (\vec{D} \times \vec{C}) = b \cdot -\vec{A} \cdot (\vec{B} \times \vec{C})$$

$b = \frac{\vec{A} \cdot (\vec{D} \times \vec{C})}{\vec{A} \cdot (\vec{B} \times \vec{C})}$	ANSWER 4.2
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for C Multiplying with $\cdot (\vec{A} \times \vec{B})$

$$\vec{D} \cdot (\vec{A} \times \vec{B}) = \underbrace{a \vec{A} \cdot (\vec{A} \times \vec{B})}_0 + \underbrace{b \vec{B} \cdot (\vec{A} \times \vec{B})}_0 + c \vec{C} \cdot (\vec{A} \times \vec{B})$$

$$\vec{D} \cdot (\vec{A} \times \vec{B}) = c \vec{C} \cdot (\vec{A} \times \vec{B})$$

$$\parallel$$

$$\vec{A} \cdot (\vec{B} \times \vec{D}) = c \vec{A} \cdot (\vec{B} \times \vec{C})$$

$$c = \frac{\vec{A} \cdot (\vec{B} \times \vec{D})}{\vec{A} \cdot (\vec{B} \times \vec{C})} \quad \text{ANSWER 4.3}$$

Problem 5

~~$$\nabla(uv) = \frac{\partial(uv)}{\partial x} \hat{x} + \frac{\partial(uv)}{\partial y} \hat{y} + \frac{\partial(uv)}{\partial z} \hat{z}$$~~

[applying product rule:

$$\frac{\partial(uv)}{\partial x} = u \frac{\partial v}{\partial x} + v \frac{\partial u}{\partial x} \quad \frac{\partial(uv)}{\partial y} = u \frac{\partial v}{\partial y} + v \frac{\partial u}{\partial y} \quad \frac{\partial(uv)}{\partial z} = u \frac{\partial v}{\partial z} + v \frac{\partial u}{\partial z}$$

arranging u and v

$$u \cdot \left(\frac{\partial v}{\partial x} \hat{x} + \frac{\partial v}{\partial y} \hat{y} + \frac{\partial v}{\partial z} \hat{z} \right) + v \cdot \left(\frac{\partial u}{\partial x} \hat{x} + \frac{\partial u}{\partial y} \hat{y} + \frac{\partial u}{\partial z} \hat{z} \right)$$

$\nabla v \quad \nabla u$

$$= u \nabla v + v \nabla u \quad \text{ANSWER 5}$$

Problem 6

$$\vec{a} \times \vec{b} = \hat{x}(a_y b_z - a_z b_y) + \hat{y}(a_z b_x - a_x b_z) + \hat{z}(a_x b_y - a_y b_x)$$

$$\vec{\nabla} \cdot (\vec{a} \times \vec{b}) = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right) \cdot (\vec{a} \times \vec{b}) =$$

$$\frac{\partial(a_y b_z - a_z b_y)}{\partial x} + \frac{\partial(a_z b_x - a_x b_z)}{\partial y} + \frac{\partial(a_x b_y - a_y b_x)}{\partial z}$$

$$\frac{\partial(a_y b_z - a_z b_y)}{\partial x} = \frac{\partial a_y b_z}{\partial x} - \frac{\partial a_z b_y}{\partial x} = b_z \frac{\partial a_y}{\partial x} + a_y \frac{\partial b_z}{\partial x} - b_y \frac{\partial a_z}{\partial x} - a_z \frac{\partial b_y}{\partial x}$$

$$\frac{\partial(a_z b_x - a_x b_z)}{\partial y} = \frac{\partial a_z b_x}{\partial y} - \frac{\partial a_x b_z}{\partial y} = b_x \frac{\partial a_z}{\partial y} + a_z \frac{\partial b_x}{\partial y} - b_z \frac{\partial a_x}{\partial y} - a_x \frac{\partial b_z}{\partial y}$$

$$\frac{\partial(a_x b_y - a_y b_x)}{\partial z} = \frac{\partial a_x b_y}{\partial z} - \frac{\partial a_y b_x}{\partial z} = b_y \frac{\partial a_x}{\partial z} + a_x \frac{\partial b_y}{\partial z} - b_x \frac{\partial a_y}{\partial z} - a_y \frac{\partial b_x}{\partial z}$$

derivatives acting on $\vec{a}$

$$b_x \left(\frac{\partial a_z}{\partial y} - \frac{\partial a_y}{\partial z} \right) + b_y \left(\frac{\partial a_x}{\partial z} - \frac{\partial a_z}{\partial x} \right) + b_z \left(\frac{\partial a_y}{\partial x} - \frac{\partial a_x}{\partial y} \right)$$

x component of
a "vector"

y component of
a "vector"

z component of
a "vector"

then it can be written as scalar product of two vectors

$$(b_x, b_y, b_z) \cdot \left(\frac{\partial a_z}{\partial y} - \frac{\partial a_y}{\partial z}, \frac{\partial a_x}{\partial z} - \frac{\partial a_z}{\partial x}, \frac{\partial a_y}{\partial x} - \frac{\partial a_x}{\partial y} \right)$$

$$\underbrace{(b_x, b_y, b_z)}_{\vec{b}}$$

↳ definition of
 $\vec{\nabla} \times \vec{a}$

$$= \vec{b} \cdot (\vec{\nabla} \times \vec{a}) + \dots$$

other page

derivatives acting on b

$$-\left[a_x \left(\frac{\partial b_z}{\partial y} - \frac{\partial b_y}{\partial z} \right) + a_y \left(\frac{\partial b_x}{\partial z} - \frac{\partial b_z}{\partial x} \right) + a_z \left(\frac{\partial b_y}{\partial x} - \frac{\partial b_x}{\partial y} \right) \right]$$

$\underbrace{\hspace{10em}}$ x component of a "vector"
 $\underbrace{\hspace{10em}}$ y component of a "vector"
 $\underbrace{\hspace{10em}}$ z component of a "vector"

$$-\left[\underbrace{(a_x, a_y, a_z)}_{\vec{a}} \cdot \underbrace{\left(\frac{\partial b_z}{\partial y} - \frac{\partial b_y}{\partial z}, \frac{\partial b_x}{\partial z} - \frac{\partial b_z}{\partial x}, \frac{\partial b_y}{\partial x} - \frac{\partial b_x}{\partial y} \right)}_{\vec{\nabla} \times \vec{b}} \right]$$

↳ representation of $\vec{\nabla} \times \vec{b}$

$$= -\vec{a} \cdot (\vec{\nabla} \times \vec{b})$$

$$\boxed{= \vec{b} \cdot (\vec{\nabla} \times \vec{a}) - \vec{a} \cdot (\vec{\nabla} \times \vec{b})} \quad \boxed{\text{ANSWER 6}}$$

Problem 7

$$(\vec{m} \times \vec{r}) = \hat{x}(m_y z - m_z y) + \hat{y}(m_z x - m_x z) + \hat{z}(m_x y - m_y x)$$

$$A_x = \frac{\mu_0}{4\pi} (m_y z - m_z y) r^{-3}, \quad A_y = \frac{\mu_0}{4\pi} (m_z x - m_x z) r^{-3}, \quad A_z = (m_x y - m_y x) r^{-3}$$

$$\vec{\nabla} \times \vec{A} = \hat{x} \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) + \hat{y} \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) + \hat{z} \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right)$$

$$\frac{\partial A_z}{\partial y} = \frac{\partial}{\partial y} \left(\frac{m_x y - m_y x}{r^3} \right) = \frac{\partial m_x y - m_y x}{\partial y} \cdot r^{-3} + m_x y - m_y x \cdot \frac{\partial r^{-3}}{\partial y}$$

$$= m_x r^{-3} + (m_x y - m_y x) \left(-\frac{3y}{r^5} \right)$$

$$\frac{\partial A_y}{\partial z} = \frac{\partial (m_z x - m_x z) r^{-3}}{\partial z} = \frac{\partial (m_z x - m_x z)}{\partial z} r^{-3} + (m_z x - m_x z) \frac{\partial r^{-3}}{\partial z}$$

$$-m_x r^{-3} + (m_z x - m_x z) \left(-\frac{3z}{r^5} \right)$$

$$m_x r^{-3} + (m_x y - m_y x) \left(-\frac{3y}{r^5} \right) - \left(-m_x r^{-3} + (m_z x - m_x z) \left(-\frac{3z}{r^5} \right) \right)$$

$$2 m_x r^{-3} + m_x y \frac{3y}{r^5} + m_y x \frac{3y}{r^5} + m_z x \frac{3z}{r^5} - m_x z \frac{3z}{r^5}$$

$$B_x = \frac{K}{r^5} \left(2 m_x r^2 - 3 m_x y^2 - 3 m_x z^2 + 3 m_y x y + 3 m_z x z \right)$$

$$\downarrow$$

$$2 m_x (x^2 + y^2 + z^2)$$

$$2 m_x x^2 + 2 m_x y^2 + 2 m_x z^2 +$$

$$= \frac{K}{r^5} \left[m_x \left(2x^2 - y^2 - z^2 \right) + 3x m_y y + 3x m_z z \right]$$

$$\downarrow$$

$$3x^2 - r^2$$

$$= \frac{K}{r^5} \left[3x^2 m_x + 3x y m_y + 3x z m_z - r^2 m_x \right]$$

$$\frac{K}{r^5} \left[3x \cdot \left(x m_x + y m_y + z m_z \right) - r^2 m_x \right]$$

$$r \cdot m$$

$$B_x = \left[\frac{3x (r \cdot m)}{r^5} - \frac{m_x}{r^3} \right] \cdot K$$

$$\mathbf{x} = r \hat{\mathbf{r}}_x, \quad \vec{r} \cdot \vec{m} = r(\hat{\mathbf{r}} \cdot \vec{m})$$

$$B_x = k \left[\frac{3 r \hat{\mathbf{r}}_x \cdot r(\hat{\mathbf{r}} \cdot \vec{m})}{r^5} - \frac{m_x}{r^3} \right]$$

$$= k \left[\frac{3 \hat{\mathbf{r}}_x (\hat{\mathbf{r}} \cdot \vec{m}) - m_x}{r^3} \right]$$

$$k = \frac{\mu_0}{4\pi}$$

for $B_x = \frac{\mu_0}{4\pi} \cdot \frac{3 \hat{\mathbf{r}}_x (\hat{\mathbf{r}} \cdot \vec{m}) - m_x}{r^3}$ if we generalize $\vec{B}$

$$= \frac{\mu_0}{4\pi} \cdot \frac{3 \hat{\mathbf{r}} (\hat{\mathbf{r}} \cdot \vec{m}) - \vec{m}}{r^3}$$

ANSWER 7

Problem 8

a) $\nabla \cdot (r^{2n} \vec{r}) = \underbrace{\nabla(r^{2n}) \cdot \vec{r}}_1 + r^{2n} \underbrace{(\nabla \cdot \vec{r})}_3 \quad \hat{\mathbf{r}} = \frac{\vec{r}}{r}$

$$2n r^{2n-1} \underbrace{\hat{\mathbf{r}} \cdot \vec{r}}_r$$

$$= 2n r^{2n-2} \underbrace{\vec{r} \cdot \vec{r}}_{r^2} + 3 r^{2n}$$

$$= 2n r^{2n} + 3 r^{2n}$$

$$= r^{2n} (2n+3)$$

ANSWER 8.a

$$b). \nabla \times (f \vec{r}) = \nabla f \times \vec{r} + f \underbrace{(\nabla \times \vec{r})}_0$$

$$\underbrace{2n r^{2n+1} \hat{r}}_0 \times \vec{r}$$

$$0 + 0 = \boxed{0} \quad \text{ANSWER 8.b}$$

$$c) \nabla \phi(r) = \frac{\partial \phi(r)}{\partial r} \hat{r} = \frac{\partial \phi(r)}{\partial r} \frac{\vec{r}}{r} = -F = -(r^{2n} \vec{r})$$

$$n \neq -1$$

$$\phi'(r) = -r^{2n+1}$$

$$\phi(r) = -\frac{r^{(2n+2)}}{2n+2} + C = \boxed{\frac{(x^2 + y^2 + z^2)^{n+1}}{2(n+1)} + C}$$

ANSWER 8.c

$$n = -1$$

$$\phi'(r) = r^{-1}, \quad \phi(r) = -\ln r + C = \boxed{-\frac{1}{2} \ln(x^2 + y^2 + z^2) + C}$$

$$d) \quad n > -1. \quad 2n+2 > 0 \quad r \rightarrow 0 \quad \phi \rightarrow 0, \quad r \rightarrow \infty \quad \phi \rightarrow \infty$$

$$n < -1 \quad 2n+2 < 0 \quad r \rightarrow 0 \quad \phi \rightarrow \infty, \quad r \rightarrow \infty \quad \phi \rightarrow 0$$

$$n = -1 \quad 2n+2 = 0 \quad r \rightarrow 0, \quad \ln(r) \rightarrow -\infty, \quad \phi \rightarrow \infty$$

$$r \rightarrow \infty, \quad \ln(r) \rightarrow \infty, \quad \phi \rightarrow -\infty$$